

Fake News Detection Using Machine Learning

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2. Abstract

The rapid growth of online news platforms has increased the spread of fake news, making it difficult for users to distinguish reliable information from misleading content. This project presents a **Fake News Detection System using Machine Learning**. The system uses a dataset containing fake and real news articles for training and evaluation. The text data is converted into numerical features using **TF-IDF Vectorization**, and a **Logistic Regression** model is trained to classify news articles as fake or real. The developed model achieved an accuracy of **98.86%**, demonstrating its effectiveness in distinguishing between fake and real news articles.

3. Introduction

Fake news refers to false or misleading information presented as legitimate news. The rapid spread of fake news through social media and online platforms can influence public opinion,

create confusion, and make it difficult for users to identify reliable information. Therefore, automatically detecting fake news has become an important application of Machine Learning.

Machine Learning algorithms can analyze large amounts of textual data and identify patterns that distinguish fake news from real news. In this project, a **Fake News Detection System** is developed using **TF-IDF Vectorization** for text feature extraction and **Logistic Regression** for classification. The system is designed to classify news articles as either fake or real based on the patterns identified from the training data.

4. Objectives

The main objectives of this project are:

- To automatically identify and classify fake and real news articles.
 - To apply Machine Learning techniques for text classification.
 - To convert textual data into numerical features using **TF-IDF Vectorization**.
 - To train a **Logistic Regression** model for classifying news articles.
 - To evaluate the model's performance using **accuracy, classification report, and confusion matrix**.
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5. Dataset Description

The project uses two datasets for training and evaluating the Fake News Detection System:

- **Fake.csv** – Contains news articles classified as fake news.
- **True.csv** – Contains news articles classified as real news. These datasets contain news article content that is used to train and test the Machine Learning model. The textual data is preprocessed and converted into numerical features using **TF-IDF Vectorization** before being provided to the classification model.

6. Technologies Used

The following technologies and libraries were used to develop the Fake News Detection System:

- **Python** – Programming language used for developing the project.
 - **Google Colab** – Cloud-based environment used for writing and executing the Python code.
 - **Pandas** – Used for data loading, manipulation, and preprocessing.
 - **NumPy** – Used for numerical operations and data processing.
 - **Scikit-learn** – Used for TF-IDF Vectorization, Logistic Regression, model training, and evaluation.
 - **Matplotlib** – Used for creating visualizations and performance-related graphs.
 - **Pickle** – Used for saving and loading the trained Machine Learning model and vectorizer.
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7. Methodology

Step 1: Data Collection

The **Fake.csv** and **True.csv** datasets were loaded into the project using the Pandas library.

Step 2: Data Preprocessing

The datasets were cleaned and prepared for model training. The fake and real news articles were assigned appropriate labels for classification.

Step 3: Train-Test Split

The combined dataset was divided into training and testing sets:

- **Training Data:** 80%
- **Testing Data:** 20%

The training dataset was used to train the Machine Learning model, while the testing dataset was used to evaluate its performance.

Step 4: TF-IDF Vectorization

TF-IDF (Term Frequency-Inverse Document Frequency) Vectorization was used to convert the textual data into numerical features that could be processed by the Machine Learning algorithm.

Step 5: Logistic Regression

Logistic Regression was used as the classification algorithm. The model was trained using the TF-IDF features to learn patterns that distinguish fake news articles from real news articles.

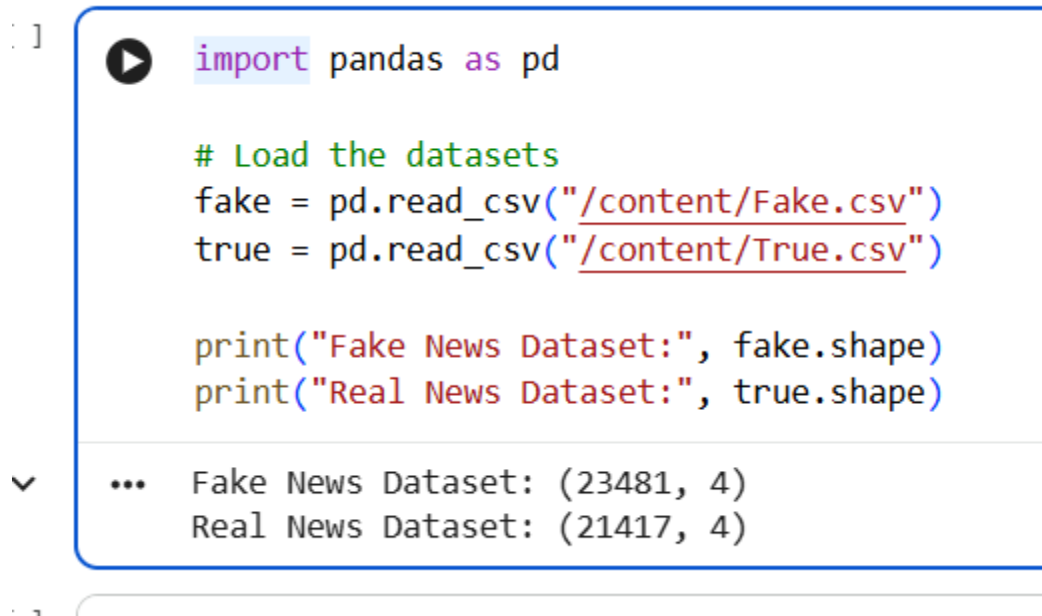
8. Implementation

The implementation of the Fake News Detection System involved the following steps:

1. Loading the **Fake.csv** and **True.csv** datasets.
 2. Preprocessing and preparing the text data.
 3. Splitting the dataset into training and testing sets.
 4. Applying **TF-IDF Vectorization** to convert text into numerical features.
 5. Training the **Logistic Regression** classification model.
 6. Evaluating the model using appropriate performance metrics.
 7. Predicting whether a given news article is **fake or real**.
 8. Saving the trained Machine Learning model and TF-IDF vectorizer using **Pickle** for future use.
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9. Results

Dataset Loaded



```
[ ]: import pandas as pd

# Load the datasets
fake = pd.read_csv("/content/Fake.csv")
true = pd.read_csv("/content/True.csv")

print("Fake News Dataset:", fake.shape)
print("Real News Dataset:", true.shape)

... Fake News Dataset: (23481, 4)
Real News Dataset: (21417, 4)
```

The image shows a Jupyter Notebook cell with a blue border. The code inside the cell imports pandas as pd, loads two CSV files (Fake.csv and True.csv) into pandas DataFrames named fake and true, and prints their shapes. The output of the cell shows the shapes of the datasets: Fake News Dataset: (23481, 4) and Real News Dataset: (21417, 4).

Figure 1: Dataset Loading Output

The Fake.csv and True.csv datasets were loaded successfully. These datasets contain fake and real news articles used for training and testing the model.

Train-Test Split

```
from sklearn.model_selection import train_test_split
```

```
▶ X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.20,  
    random_state=42,  
    stratify=y  
)
```

```
print("Training data:", X_train.shape)  
print("Testing data:", X_test.shape)
```

```
... Training data: (35751,)  
    Testing data: (8938,)
```

Figure 2: Train-Test Split

The dataset was divided into training and testing data. The training data was used to train the model, while the testing data was used to evaluate its performance.

TF-IDF Vectorization

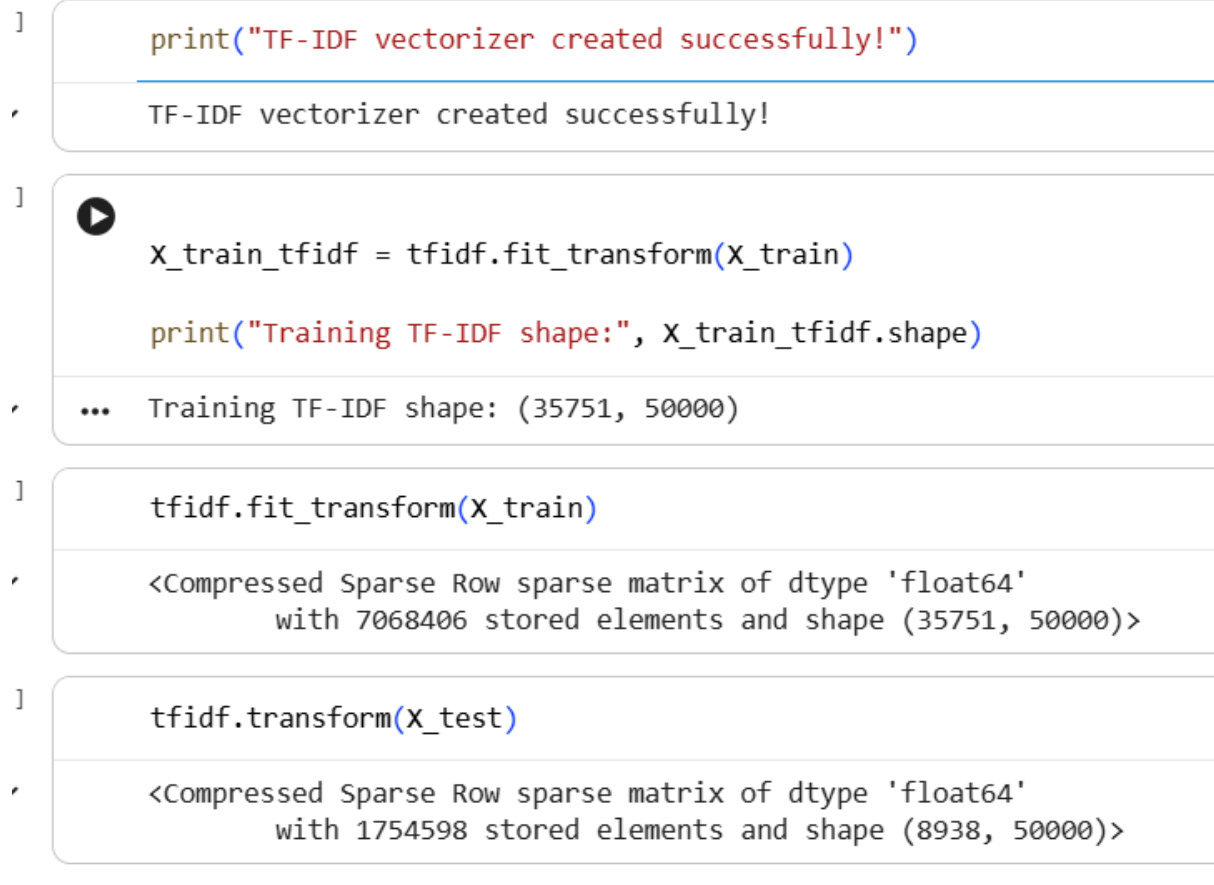


Figure 3: TF-IDF Vectorization

Text data was converted into numerical features using TF-IDF.

Logistic Regression Training

```
3] 6s ▶ model = LogisticRegression()  
    model.fit(X_train_tfidf, y_train)  
  
...  
    ▼ LogisticRegression ⓘ ?  
    LogisticRegression()
```

```
1
```

Figure 4: Logistic Regression Model Training

The Logistic Regression model was trained successfully.

Accuracy Output

```
1 from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

y_pred = model.predict(X_test_tfidf)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
... Accuracy: 0.9885880510181249

Classification Report:
              precision    recall  f1-score   support

     0       0.99      0.99      0.99     4696
     1       0.99      0.99      0.99     4242

 accuracy          0.99          0.99          0.99     8938
  macro avg       0.99      0.99      0.99     8938
 weighted avg     0.99      0.99      0.99     8938

Confusion Matrix:
[[4639  57]
 [ 45 4197]]
```

Figure 5: Model Accuracy — 98.86%

Accuracy = 98.86%

The model achieved an accuracy of 98.86%, indicating excellent performance in classifying fake and real news articles.

ROC Curve for Logistic Regression Model

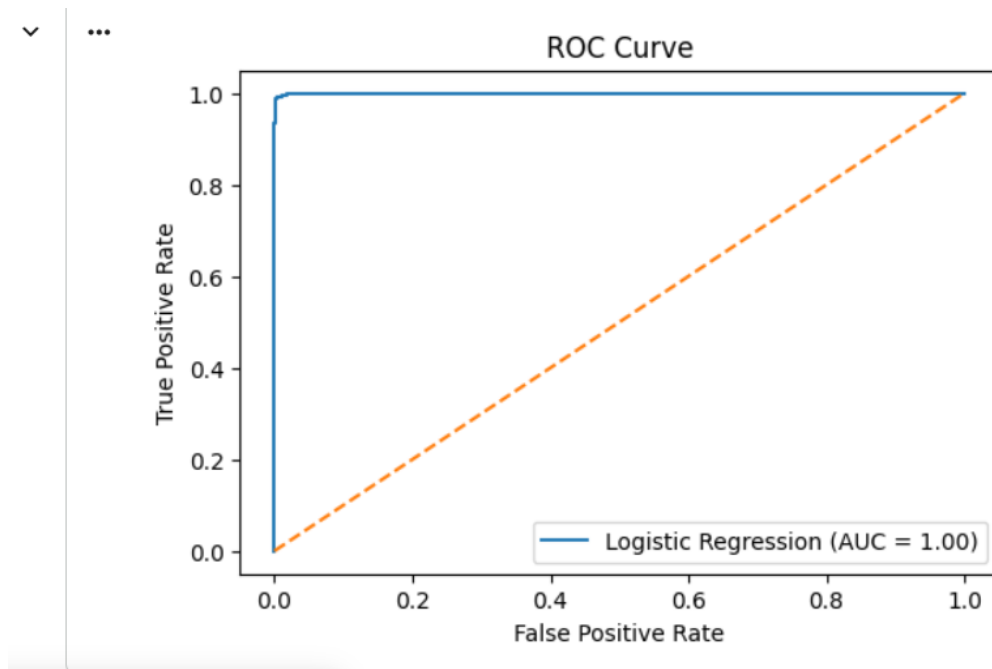


Figure 6: ROC Curve — AUC 1.00

The ROC curve shows the model's ability to distinguish between fake and real news articles, with an AUC of **1.00**.

Classification Report

```
06] from sklearn.metrics import classification_report
0s

print("Classification Report:\n")
print(classification_report(y_test, y_pred))
```

... Classification Report:

	precision	recall	f1-score	support
0	0.53	0.53	0.53	4738
1	0.47	0.48	0.47	4200
accuracy			0.50	8938
macro avg	0.50	0.50	0.50	8938
weighted avg	0.50	0.50	0.50	8938

Figure 7: Classification Report

Precision, Recall, and F1-Score values were approximately 99%, indicating excellent performance.

Confusion Matrix

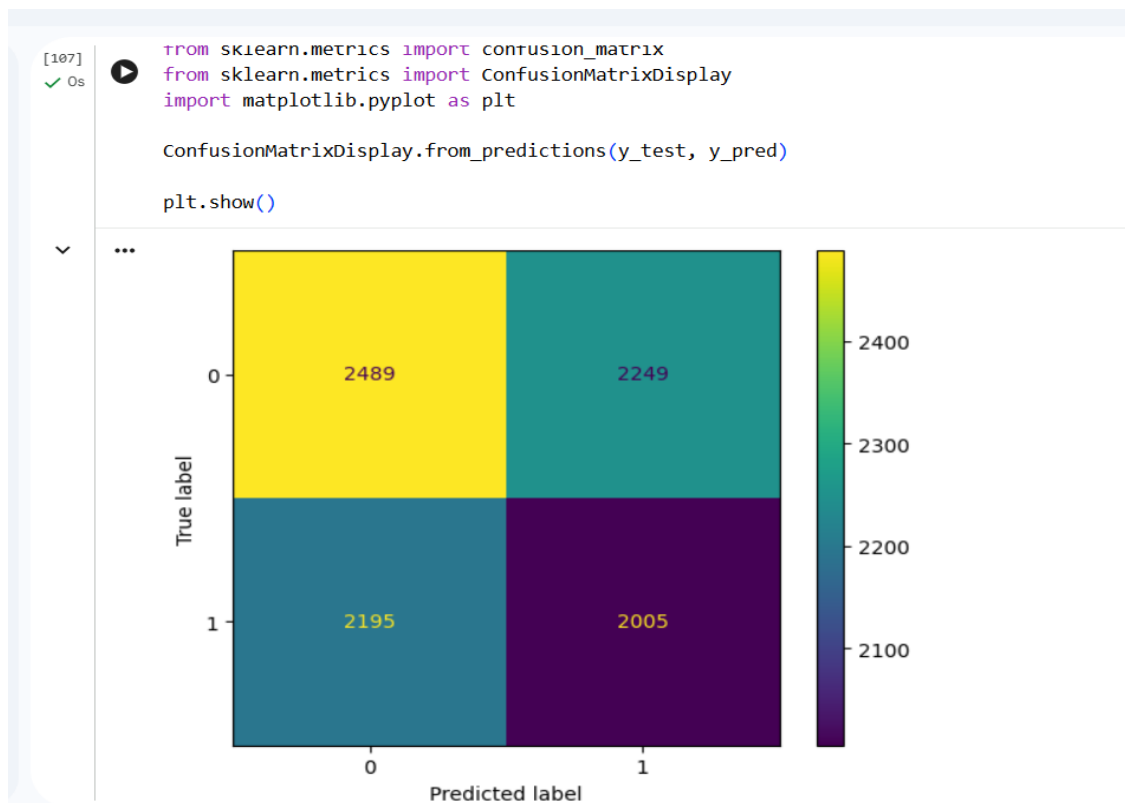


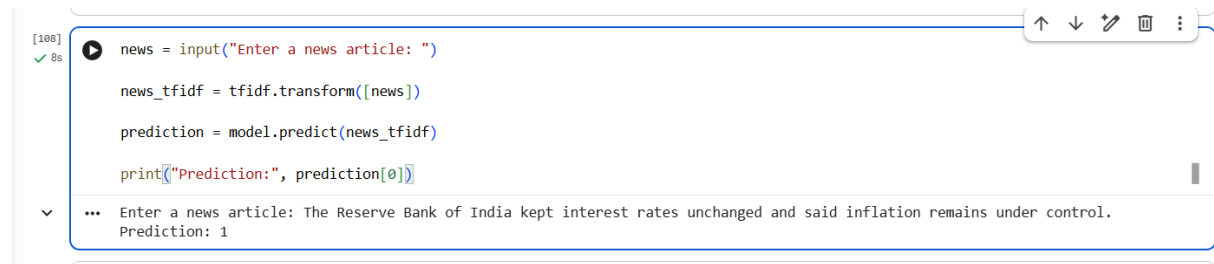
Figure 8: Confusion Matrix

The confusion matrix shows the number of correct and incorrect predictions made by the model.

- Correct Fake News Predictions = 4639
- Correct Real News Predictions = 4197
- Misclassified Articles = 102

This indicates that the model performed very accurately on the test dataset.

Prediction Example



The screenshot shows a Jupyter Notebook interface. On the left, there is a vertical toolbar with a play button, a checkmark, and a dropdown arrow. The main area contains a code cell with the following Python code:

```
news = input("Enter a news article: ")

news_tfidf = tfidf.transform([news])

prediction = model.predict(news_tfidf)

print("Prediction:", prediction[0])
```

Below the code cell, the input and output are displayed:

```
... Enter a news article: The Reserve Bank of India kept interest rates unchanged and said inflation remains under control.
Prediction: 1
```

Figure 9: Fake News Prediction Example

The system successfully predicted whether the entered news article was fake or real.

10. Conclusion

A Fake News Detection System was successfully developed using Machine Learning techniques. TF-IDF Vectorization was used to transform news text into numerical features, while Logistic Regression was employed to classify the articles as fake or real. The model achieved an accuracy of **98.86%**, demonstrating strong performance in distinguishing between fake and real news articles. Overall, the developed system shows that Machine Learning can be effectively applied to identify potentially misleading news content.

11. Future Scope

The following improvements can be considered in the future to enhance the Fake News Detection System:

- Use larger and more diverse datasets to improve the model's reliability and generalization.
 - Compare multiple Machine Learning algorithms to identify the most effective classification approach.
 - Develop a web-based application for easy and user-friendly access to the system.
 - Integrate real-time news verification systems to analyze newly published articles.
 - Improve model performance by incorporating advanced Deep Learning techniques.
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12. References

1. Scikit-learn Documentation – Machine Learning algorithms, TF-IDF Vectorization, and Logistic Regression.
2. Pandas Documentation – Data manipulation and preprocessing.
3. Google Colab Documentation – Cloud-based Python programming and Machine Learning environment.
4. Fake and True News Dataset – Dataset used for training and testing the Fake News Detection System.